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MEMORY STRUCTURE HAVING AIR GAP AND METHOD FOR MANUFACTURING THE SAME

Abstract

A semiconductor device includes a substrate and a memory structure disposed over the substrate. The memory structure includes a pair of first conductive lines, a channel element disposed between the pair of the first conductive lines and formed with an air gap therein, a first memory element disposed to separate one of the pair of the first conductive lines from the channel element, and a second memory element disposed to separate the other one of the pair of the first conductive lines from the channel element. A method for manufacturing the semiconductor device is also disclosed.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE [0001] This application is a continuation of U.S. patent application Ser. No. 17/707,562, filed on Mar. 29, 2022, all of which are hereby expressly incorporated by reference into the present application.

BACKGROUND
[0002] Semiconductor memory structures are widely used in computers, portable devices, automotive parts, and internet of things (IoT), etc. With increasing requirement of the semiconductor memory structures to have a high memory capacity, in addition to scale down memory cells, a memory array tends to be developed to have a three-dimensional (3D) architecture instead of a two-dimensional (2D) architecture, so that the memory capacity of the semiconductor memory structures can be effectively increased with a relatively small area penalty.
[0003] A capacitance between two adjacent ones of a plurality of word lines of a 3D semiconductor memory structure is very important to determine an operation speed of the 3D semiconductor memory structure. Hence, there are demands to develop a 3D semiconductor memory structure with a reduced capacitance between two adjacent ones of the word lines.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
[0005] FIG. **1** is a flow diagram illustrating a method for manufacturing a semiconductor device in accordance with some embodiments.
[0006] FIGS. **2**A to **16**B are schematic perspective views illustrating some intermediate stages of the manufacturing method as depicted in FIG. **1** in accordance with some embodiments.
[0007] FIG. **17**A is a schematic perspective view illustrating a portion of the semiconductor device shown in FIG. **16**A.
[0008] FIG. **17**B is a schematic sectional view of the portion of the semiconductor device taken along line A-A of FIG. **17**A.
[0009] FIG. **17**C is another schematic sectional view of the portion of the semiconductor device taken along line B-B of FIG. **17**A.
[0010] FIG. **18**A is a schematic perspective view illustrating another portion of the semiconductor device shown in FIG. **16**A.
[0011] FIG. **18**B is a schematic sectional view of the another portion of the semiconductor device taken along line C-C of FIG. **18**A.
[0012] FIG. **18**C is another schematic sectional view of the another portion of the semiconductor device taken along line D-D of FIG. **18**A.
[0013] FIG. **19** is a schematic sectional view illustrating a portion of a semiconductor device in accordance with some alternative embodiments.
[0014] FIG. **20** is a schematic sectional view illustrating a portion of a semiconductor device in

accordance with some alternative embodiments.
[0015] FIG. **21** is a schematic sectional view illustrating a portion of a semiconductor device in accordance with some alternative embodiments.
[0016] FIG. **22** is a schematic sectional view illustrating a portion of a semiconductor device in accordance with some alternative embodiments.
[0017] FIG. **23** is a schematic sectional view illustrating the operation of the semiconductor device in accordance with some embodiments.

DETAILED DESCRIPTION

[0018] The following disclosure provides many different embodiments, or examples, for implementing different features of the disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
[0019] Further, spatially relative terms, such as “on,” “above,” “upper,” “lower,” “over,” “downwardly,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
[0020] The present disclosure is directed to a semiconductor device including a three-dimensional (3D) memory structure with air gaps and a method for manufacturing the same. FIG. **1** illustrates a method **100** for manufacturing a semiconductor device in accordance with some embodiments. FIGS. **2**A to **16**B illustrate schematic perspective views of some intermediate stages of the method **100** shown in FIG. **1**. Additional steps can be provided before, after or during the method **100**, and some of the steps described herein may be replaced by other steps or be eliminated. Similarly, further additional features may be present in a semiconductor device **200** illustrated in FIG. **16**A, and/or features present may be replaced or eliminated in additional embodiments.
[0021] Referring to FIG. **1** and the example illustrated in FIGS. **2**A and **2**B, the method **100** begins at step **101**, where a stack assembly is formed over a substrate. FIG. **2**A is a schematic perspective view of a stack assembly **2** formed over a substrate **1**. The stack assembly **2** includes an etch stop layer (ESL) **21**, a multi-layer dielectric stack **22**, and a dummy layer **23**, which are formed sequentially over the substrate **1**. FIG. **2**B is a schematic perspective view of a portion of the configuration shown in FIG. **2**A.
[0022] In some embodiments, the substrate **1** may include a semiconductor substrate. The semiconductor substrate may be, for example, but not limited to, a silicon-on-insulator (SOI) substrate, a germanium-on-insulator (GOI) substrate, a bulk semiconductor substrate, or the like, and may be doped with a dopant. The substrate **1** may have multiple layers. The substrate **1** may include elemental semiconductor materials, such as crystalline silicon, diamond, or germanium; compound semiconductor materials, such as silicon carbide, gallium arsenic, indium arsenide, gallium phosphide, indium arsenide, indium phosphide, or indium antimonide; alloy semiconductor materials, such as silicon germanium, silicon germanium carbide, gallium arsenic phosphide, aluminum gallium arsenide, or gallium indium phosphide; or combinations thereof. Other materials suitable for the substrate **1** are within the contemplated scope of the disclosure.

[0023] The etch stop layer **21** is formed over the substrate **1** by a suitable fabrication technique known to those skilled in the art of semiconductor fabrication, for example, but not limited to, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), plasma-enhanced CVD (PECVD), or the like. Other suitable techniques for forming the etch stop layer **21** are within the contemplated scope of the disclosure. In some embodiments, the etch stop layer **21** may be made of a dielectric material, for example, but not limited to, silicon nitride, silicon nitride doped with carbon, silicon oxide, silicon oxynitride, silicon oxynitride doped with carbon, amorphous carbon material, silicon carbide, other nitride materials, other carbide materials, aluminum oxide, other oxide materials, other metal oxides, boron nitride, boron carbide, and other low-k dielectric materials or low-k dielectric materials doped with one or more of carbon, nitrogen, and hydrogen, or other suitable materials. Other materials suitable for the etch stop layer **21** are within the contemplated scope of the disclosure.

[0024] The multi-layer dielectric stack **22** includes a plurality of first dielectric layers **221** and a plurality of second dielectric layers **222**, which are alternately stacked on the etch stop layer **21**. In some embodiments, the first dielectric layers **221** may include, for example, but not limited to, silicon oxide (SiO), and the second dielectric layers **222** may include, for example, but not limited to, silicon nitride (SiN). Other materials suitable for the first and second dielectric layers **221**, **222** are within the contemplated scope of the disclosure. In some embodiments, the uppermost and lowermost layers of the multi-layer dielectric stack **22** are the second dielectric layers **222**, and are disposed to be in contact with the dummy layer **23** and the etch stop layer **21**, respectively.

[0025] The dummy layer **23** may be made of a material having an etching selectivity different from those of the materials for the first and second dielectric layers **221**, **222**. In some embodiments, the material for the dummy layer **23** may include, for example, but not limited to, polysilicon, silicon carbide, silicon oxycarbide, or the like, or combinations thereof. Other materials suitable for the dummy layer **23** are within the contemplated scope of the disclosure. The dummy layer **23** may be formed on the multi-layer dielectric stack **22** by a suitable fabrication technique known to those skilled in the art of semiconductor fabrication, for example, but not limited to, CVD, PVD, ALD, PECVD, or the like. Other suitable techniques for forming the dummy layer **23** are within the contemplated scope of the disclosure.

[0026] Referring to FIG. **1** and the example illustrated in FIGS. **3**A and **3**B, the method **100** then proceeds to step **102**, where a plurality of first elongated slots are formed. FIG. **3**A illustrates a configuration subsequent to that shown in FIG. **2**A, and FIG. **3**B is a schematic perspective view of a portion of the configuration shown in FIG. **3**A. In some embodiments, formation of the first elongated slots **223** in an array arrangement includes the following steps. First, a hard mask (not shown) is formed on the dummy layer **23** shown in FIG. **2**A by a suitable fabrication technique known to those skilled in the art of semiconductor fabrication, for example, but not limited to, CVD, PVD, ALD, PECVD, or the like. Other suitable techniques for forming the hard mask are within the contemplated scope of the disclosure. A photoresist layer (not shown) is then formed on the hard mask by a suitable fabrication technique known to those skilled in the art of semiconductor fabrication, for example, but not limited to, a spin-on technique. Other suitable techniques for forming the photoresist layer are within the contemplated scope of the disclosure. The photoresist layer is then patterned using a suitable photolithography technique to form a pattern of recesses in position corresponding to the first elongated slots **223** to be formed. For example, the photoresist layer is exposed to light for patterning, followed by developing to form the pattern of the recesses. The pattern of the recesses formed in the photoresist layer is transferred to the hard mask using a suitable etching process, for example, but not limited to, a wet etching process, a dry etching process, a reactive ion etching process, a neutral beam etching process, or the like. After the pattern of the recesses is transferred to the hard mask, the photoresist layer may be removed by, for example, but not limited to, an ashing process. The pattern of the recesses formed in the hard mask is then transferred to the dummy layer **23** using a suitable etching process, for example, but not limited to, a wet etching process, a dry etching process, or the like, so as to form an array of openings **231** extending through the dummy layer **23**. Thereafter, the hard mask may be removed by a suitable process, for example, but not limited to, a wet etching process, a dry etching process, a planarization process, or the like. An array of the openings **231** formed in the dummy layer **23** includes a plurality of columns of the openings **231** spaced apart from each other in a first direction (e.g., an X direction direction), and the openings **231** in each of the columns are spaced apart from each other in a second direction (e.g., a Y direction) transverse to the first direction. The pattern of the openings **231** formed in the dummy layer **23** is then transferred to the multi-layer dielectric stack **22** using a suitable etching process, for example, but not limited to, a wet etching process, a dry etching process, or the like, so as to form an array of the first elongated slots **223** extending downwardly from an upper surface of the dummy layer **23** and through the multi-layer dielectric stack **22** in a third direction (e.g., a Z direction) transverse to the first and second directions to terminate at the etch stop layer **21**. In some embodiments, the first, second, and third directions are perpendicular to one another. In some embodiments, the hard mask may be removed after the first elongated slots **223** are formed. An array of the first elongated slots **223** includes a plurality of columns of the first elongated slots **223** spaced apart from each other in the X direction, and the first elongated slots **223** in each of the columns are spaced part from each other in the Y direction. Each of the first dielectric layers **221** and the second dielectric layers **222** is thus patterned into a grid-shaped configuration. Each of the first dielectric layers **221** in the grid-shaped configuration includes a plurality of first dielectric portions **2211** elongated in the Y direction and spaced apart from each other in the X direction, and a plurality of columns of second dielectric portions **2212** disposed to alternate with the first dielectric portions **2211** in the X direction. The columns of the second dielectric portions **2212** are spaced apart from each other in the X direction, and the second dielectric portions **2212** in each of the columns are spaced apart from each other in the Y direction. Two adjacent ones of the first dielectric portions **2211** are interconnected with each other through a corresponding one column of the second dielectric portions **2212**. Similarly, each of the second dielectric layers **222** in the grid-shaped configuration includes a plurality of third dielectric portions **2221** elongated in the Y direction and spaced apart from each other in the X direction, and a plurality of columns of fourth dielectric portions **2222** disposed to alternate with the third dielectric portions **2221** in the X direction. The columns of the fourth dielectric portions **2222** are spaced apart from each other in the X direction, and the fourth dielectric portions **2222** in each of the columns are spaced apart from each other in the Y direction. Two adjacent ones of the third dielectric portions **2221** are interconnected with each other through a corresponding one column of the fourth dielectric portions **2222**. Similarly, the dummy layer **23** is patterned into the grid-shaped configuration, and includes a plurality of first dummy portions **232** elongated in the Y direction and spaced apart from each other in the X direction and a plurality of columns of second dummy portions **233** disposed to alternate with the first dummy portions **232** in the X direction. The columns of the second dummy portions **233** are spaced apart from each other in the X direction, and the second dummy portions **233** in each of the columns are spaced apart from each other in the Y direction. Two adjacent ones of the first dummy portions **233** are interconnected with each other through a corresponding one column of the second dummy portions **233**.

[0027] Referring to FIG. **1** and the example illustrated in FIGS. **4**A and **4**B, the method **100** then proceeds to step **103**, where a plurality of first dielectric bars are formed. FIG. **4**A illustrates a configuration subsequent to that shown in FIG. **3**A, and FIG. **4**B is a schematic perspective view of a portion of the configuration shown in FIG. **4**A. The first elongated slots **223** shown in FIG. **3**A are filled with a dielectric material, and excess of the dielectric material above the dummy layer **23** is removed by a planarization technique, such as chemical mechanical planarization (CMP) such that a plurality of first dielectric bars **224** in an array arrangement are formed and are elongated in the Z direction. An array of the first dielectric bars **224** includes a plurality of columns of the first dielectric bars **224** spaced apart from each other in the X direction, and the first dielectric bars **224** in each of the columns are spaced apart from each other in the Y direction. In some embodiments, the dielectric material for forming the first dielectric bars **224** may be the same as a dielectric material for forming the second dielectric layers **222**. In some embodiments, both the dielectric material for forming the first dielectric bars **224** and the dielectric material for forming the second dielectric layers **222** are silicon nitride (SiN).

[0028] Referring to FIG. **1** and the example illustrated in FIGS. **5**A and **5**B, the method **100** then proceeds to step **104**, where a plurality of second elongated slots are formed. FIG. **5**A illustrates a configuration subsequent to that shown in FIG. **4**A, and FIG. **5**B is a schematic perspective view of a portion of the configuration shown in FIG. **5**A. In some embodiments, the process for forming the second elongated slots **225** in an array arrangement is the same as or similar to that for forming the first elongated slots **223** described above, and the details thereof are omitted for the sake of brevity. In the formation of the second elongated slots **225**, the second and fourth dielectric portions **2212**, **2222** are removed, such that the first dummy portions **232**, the first dielectric portions **2211**, and the third dielectric portions **2221** remain and a plurality of the second elongated slots **225** in an array arrangement are formed to extend downwardly from an upper surface of the configuration shown in FIG. **5**A in the Z direction to terminate at the etch stop layer **21**. An array of the second elongated slots **225** includes a plurality of columns of the second elongated slots **225** spaced apart from each other in the X direction, and the second elongated slots **225** in each of the columns are spaced part from each other in the Y direction. The first dielectric portions **2211** are disposed in an array arrangement and are elongated in the Y direction. The array of the first dielectric portions **2211** includes a plurality of columns of the first dielectric portions **2211** spaced apart from each other in the X direction, and the first dielectric portions **2211** in each of the columns are spaced apart from each other in the Z direction. Similarly, the third dielectric portions **2221** are disposed in an array arrangement and are elongated in the Y direction. An array of the third dielectric portions **2221** includes a plurality of columns of the third dielectric portions **2221** spaced apart from each other in the X direction, and the third dielectric portions **2221** in each of the columns are spaced apart from each other in the Z direction. The first dummy portions **232** are elongated in the Y direction and are spaced apart from each other in the X direction.

[0029] Referring to FIG. **1** and the example illustrated in FIGS. **6**A and **6**B, the method **100** then proceeds to step **105**, where a plurality of dummy pillars are formed. FIG. **6**A illustrates a configuration subsequent to that shown in FIG. **5**A, and FIG. **6**B is a schematic perspective view of a portion of the configuration shown in FIG. **6**A. The second elongated slots **225** shown in FIG. **5**A are filled with a dummy material, and excess of the dummy material above the first dummy portions **232** and the first dielectric bars **224** is removed by a planarization technique, for example, CMP, such that a plurality of dummy pillars **24** in an array arrangement are formed and are elongated downwardly in the Z direction to terminate at the etch stop layer **21**. An array of the dummy pillars **24** includes a plurality of columns of the dummy pillars **24** spaced apart from each other in the X direction, and the dummy pillars **24** in each of the columns are spaced apart from each other in the Y direction. Two adjacent ones of the first dummy portions **232** are interconnected with each other through upper portions of a corresponding one column of the dummy pillars **24**. In some embodiments, the dummy material for forming the dummy pillars **24** may be the same as that for forming the first dummy portions **232** (i.e., the same as that for forming the dummy layer **23**).

[0030] Referring to FIG. **1** and the example illustrated in FIGS. **7**A and **7**B, the method **100** then proceeds to step **106**, where a plurality of third elongated slots and a plurality of fourth elongated slots are formed. FIG. **7**A illustrates a configuration subsequent to that shown in FIG. **6**A, and FIG. **7**B is a schematic perspective view of a portion of the configuration shown in FIG. **7**A. The first dielectric bars **224** and the third dielectric portions **2221** shown in FIG. **6**A are removed using a suitable etching process, for example, but not limited to, an isotropic wet etching process, an isotropic dry etching process, or the like, such that a plurality of third elongated slots **226** in an array arrangement and a plurality of fourth elongated slots **227** in an array arrangement are formed.

The third elongated slots **226** in the array arrangement extend downwardly in the Z direction to terminate at the etch stop layer **21**. An array of the third elongated slots **226** includes a plurality of columns of the third elongated slots **226** spaced apart from each other in the X direction. The third elongated slots **226** in each of the columns are spaced apart from each other in the Y direction. The fourth elongated slots **227** in the array arrangement extend in the Y direction. An array of the fourth elongated slots **227** includes a plurality of columns of the fourth elongated slots **227** spaced apart from each other in the X direction. The fourth elongated slots **227** in each of the columns are spaced apart from each other in the Z direction. The fourth elongated slots **227** are in spatial communication with the third elongated slots **226**. Two adjacent columns of the dummy pillars **24** are interconnected with each other through a corresponding one column of the first dielectric portions **2211**, each of which is configured as a dielectric pillar elongated in the Y direction. In each column of the first dielectric portions **2211**, two adjacent ones of the first dielectric portions **2211** have a first wall surface **2211***a* and a second wall surface **2211***b*, respectively, which face each other and are spaced apart from each other by a corresponding one of the fourth elongated slots **227** in the Z direction. In two adjacent columns of the dummy pillars **24** spaced apart from each other by the each column of the first dielectric portions **2211**, each of the dummy pillars **24** in one column of the dummy pillars **224** and a corresponding one of the dummy pillars **24** in the other one column of the dummy pillars **24** have a third wall surface **24***a* and a fourth wall surface **24***b*, respectively, which extend to interconnect the two adjacent ones of the first dielectric portions **2211** of the each column of the first dielectric portions **2211**.
[0031] Referring to FIG. **1** and the example illustrated in FIGS. **8**A and **8**B, the method **100** then proceeds to step **107**, where a plurality of channel features and a plurality of second dielectric bars are formed. FIG. **8**A illustrates a configuration subsequent to that shown in FIG. **7**A, and FIG. **8**B is a schematic perspective view of a portion of the configuration shown in FIG. **8**A. A channel layer is conformally deposited and a dielectric material is filled into the third and fourth elongated slots **226**, **227** shown in FIG. **7**A independently by a suitable deposition technique known to those skilled in the art of semiconductor fabrication, for example, but not limited to, CVD, ALD, PVD, PECVD, or the like. Other suitable techniques for conformally depositing the channel layer and filling the dielectric material are within the contemplated scope of the disclosure.
[0032] In some embodiments, the channel layer may be made of various semiconductor material. In some embodiments, the semiconductor material for making the channel layer may include, for example, but not limited to, polysilicon, an indium-comprising material, such as In.sub.x1Ga.sub.x2Zn.sub.x3M.sub.x4O, where M may be Ti, Al, Ag, Si, Sn, W, or the like, and x1, x2, x3 and x4 may each be any value between 0 and 1, or the like, or combinations thereof. In some embodiments, the channel layer may be formed as a single layer having one of the aforesaid materials. In some alternative embodiments, the channel layer may be formed as a laminate structure having at least two of the aforesaid materials in various constitutions. In some embodiments, the channel layer may be doped with a dopant to achieve extra stability. Other materials suitable for the channel layer are within the contemplated scope of the disclosure.
[0033] In some embodiments, the dielectric material for filling the third and fourth elongated slots **226**, **227** has an etching selectively different from that of the first dielectric portions **2211**. In some embodiments, the dielectric material for filling the third and fourth elongated slots **226**, **227** includes silicon nitride (SiN).
[0034] Portions of the channel layer and portions of the dielectric material in the third elongated slots **226** are removed using a suitable etching process, for example, but not limited to, a wet etching process, a dry etching process, or the like, to form a plurality of second dielectric bars **228** in an array arrangement and a plurality of channel features **31** in an array arrangement and to reopen the third elongated slots **226**. The second dielectric bars **228** in the array arrangement are elongated in the Y direction. An array of the second dielectric bars **228** includes a plurality of columns of the second dielectric bars **228** spaced apart from each other in the X direction. The

second dielectric bars **228** in each of the columns are disposed to alternate with the first dielectric portions **2211** of a corresponding one column of the first dielectric portions **2211** in the Z direction. The channel features **31** in the array arrangement are elongated in the Y direction. An array of the channel features **31** includes a plurality of columns of the channel features **31** spaced apart from each other in the X direction. Each of the channel features **31** includes a plurality of first channel regions **311** and a plurality of second channel regions **312** disposed to alternate with the first channel regions **311** in the Y direction. Each of the first channel regions **311** is in contact with two corresponding ones of the dummy pillars **24**, and the second channel regions **312** are exposed from the dummy pillars **24**.

[0035] Referring to FIG. **1** and the example illustrated in FIGS. **9**A and **9**B, the method **100** then proceeds to step **108**, where the first dielectric portions are removed. FIG. **9**A illustrates a configuration subsequent to that shown in FIG. **8**A, and FIG. **9**B is a schematic perspective view of a portion of the configuration shown in FIG. **9**A. The first dielectric portions **2211** shown in FIG. **8**A are removed using a suitable etching process, for example, but not limited to, a wet etching process, a dry etching process, or the like, to form a plurality of fifth elongated slots **229** in an array arrangement. The fifth elongated slots **229** in the array arrangement are elongated in the Y direction. An array of the fifth elongated slots **229** includes a plurality of columns of the fifth elongated slots **229** spaced apart from each other in the X direction. The fifth elongated slots **229** in each of the columns are spaced apart from each other in the Z direction. The fifth elongated slots **229** are in spatial communication with the third elongated slots **226**.

[0036] Referring to FIG. **1** and the example illustrated in FIGS. **10**A and **10**B, the method **100** then proceeds to step **109**, where a plurality of memory elements and a plurality of first conductive lines are formed. FIG. **10**A illustrates a configuration subsequent to that shown in FIG. **9**A, and FIG. **10**B is a schematic perspective view of a portion of the configuration shown in FIG. **10**A. A memory layer is conformally deposited and a conductive material is filled into the third elongated slots **226** and the fifth elongated slots **229** independently by a suitable deposition technique known to those skilled in the art of semiconductor fabrication, for example, but not limited to, CVD, ALD, PVD, PECVD, or the like. Other suitable techniques for conformally depositing the memory layer and filling the conductive material are within the contemplated scope of the disclosure.

[0037] In some embodiments, the memory layer may be made of a high-k dielectric material. In some embodiments, the memory layer may include, for example, but not limited to, a ferroelectric material, silicon nitride, silicon oxynitride, silicon oxide, or the like. The ferroelectric material may be binary oxides such as hafnium oxide (hafnia, HfO.sub.2), ternary oxides such as hafnium silicate (HfSiO.sub.x), hafnium zirconate (HfZrO.sub.x), barium titanate (BaTiO.sub.3), lead titanate (PbTiO.sub.3), strontium titanate (SrTiO.sub.3), calcium manganite (CaMnO.sub.3), bismuth ferrite (BiFeO.sub.3), or the like, or quaternary oxides such as barium strontium titanate (BaSrTiO.sub.x), or the like, or combinations thereof. In some embodiments, the memory layer may have a multi-layered structure. Other materials suitable for the memory layer are within the contemplated scope of the disclosure.

[0038] In some embodiments, the conductive material may include, for example, but not limited to, aluminum, zirconium, titanium, tungsten, tantalum, ruthenium, palladium, platinum, cobalt, nickel, or the like, or alloys thereof. Other materials suitable for the conductive material are within the contemplated scope of the disclosure.

[0039] Portions of the memory layer and portions of the conductive material in the third elongate slots **226** are removed using a suitable etching process, for example, but not limited to, a wet etching process, a dry etching process, or the like, to form a plurality of first conductive lines **41** in an array arrangement and a plurality of memory elements **51** in an array arrangement. The first conductive lines **41** in the array arrangement are elongated in the Y direction. An array of the first conductive lines **41** includes a plurality of columns of the conductive lines **41** spaced apart from each other in the X direction. The first conductive lines **41** in each of the columns are disposed to alternate with the second dielectric bars **228** of a corresponding one column of the second dielectric bars **228** in the Z direction. The memory elements **51** in the array arrangement are elongated in the Y direction. An array of the memory elements **51** includes a plurality of columns of the memory elements **51** spaced apart from each other in the X direction. The memory elements **51** in each of the columns and the channel features **31** in a corresponding one of the columns are disposed to alternate and in contact with each other.

[0040] Referring to FIG. **1** and the example illustrated in FIGS. **11**A and **11**B, the method **100** then proceeds to step **110**, where a plurality of channel elements are formed. FIG. **11**A illustrates a configuration subsequent to that shown in FIG. **10**A, and FIG. **11**B is a schematic perspective view of a portion of the configuration shown in FIG. **11**A. The second dielectric bars **228** and the second channel regions **312** are removed using a suitable etching process, for example, but not limited to, a wet etching process, a dry etching process, or the like, to form a plurality of channel elements **311** (i.e., the first channel regions **311**) in a three dimensional (3D) array arrangement and to reopen the third and fourth elongated slots **226**, **227**. In some embodiments, the second channel regions **312** are removed completely. In some embodiments, the second channel regions **312** are removed incompletely. In some embodiments, portions of the first channel regions **311** proximate to the second channel regions **312** may also be removed in addition to complete removal of the second channel regions **312**.

[0041] A 3D array of the channel elements **311** includes a plurality of two dimensional (2D) arrays of the channel elements **311** spaced apart from each other in the X direction. Each of the 2D arrays of the channel elements **311** includes a plurality of columns of the channel elements **311** spaced apart from each other in the Z direction, and the channel elements **311** in each of the columns of each of the 2D arrays of the channel elements **311** are spaced apart from each other in the Y direction. Each of the fourth elongated slots **227** passes through the channel elements **311** arranged in a corresponding one column. Each of the channel elements **311** is in contact with two corresponding ones of the dummy pillars **24** and two corresponding ones of the memory elements **51**.

[0042] Referring to FIG. **1** and the example illustrated in FIGS. **12**A and **12**B, the method **100** then proceeds to step **111**, where a plurality of air gaps are formed. FIG. **12**A illustrates a configuration subsequent to that shown in FIG. **11**A, and FIG. **12**B is a schematic perspective view of a portion of the configuration shown in FIG. **12**A. The third and fourth elongated slots **226**, **227** shown in FIG. **11**A are filled with a dielectric material **25** by a suitable deposition technique known to those skilled in the art of semiconductor fabrication, for example, but not limited to, CVD, ALD, PVD, PECVD, or the like, such that air gaps **32** are formed in the channel elements **311**, as shown in FIGS. **17**B, **18**B, **19**, and **20**. Other suitable techniques for filling the dielectric material **25** are within the contemplated scope of the disclosure. In some embodiments, the air gaps **32** may be additionally formed in the dielectric material **25** filled among the dummy pillars **24** and/or in the dielectric material **25** filled between two adjacent ones of the memory elements **51** of each of the columns of the memory elements **51**. In some embodiments, the dielectric material **25** for filling the third and fourth elongated slots **226**, **227** may include, for example, but not limited to, silicon nitride, silicon nitride doped with carbon, silicon oxide, silicon oxynitride, silicon oxynitride doped with carbon, amorphous carbon material, silicon carbide, other nitride materials, other carbide materials, other low-k dielectric materials or low-k dielectric materials doped with one or more of carbon, nitrogen, and hydrogen, or other suitable materials.

[0043] Referring to FIG. **1** and the example illustrated in FIGS. **13**A and **13**B, the method **100** then proceeds to step **112**, where a passivation layer is formed. FIG. **13**A illustrates a configuration subsequent to that shown in FIG. **12**A, and FIG. **13**B is a schematic perspective view of a portion of the configuration shown in FIG. **13**A. A passivation layer **61** is formed on the configuration shown in FIG. **12**A by a suitable deposition technique known to those skilled in the art of semiconductor fabrication, for example, but not limited to, CVD, ALD, PVD, PECVD, or the like.

Other suitable techniques for forming the passivation layer **61** are within the contemplated scope of the disclosure. The passivation layer **61** may include a dielectric material, for example, but not limited to, silicon nitride, silicon nitride doped with carbon, silicon oxide, silicon oxynitride, silicon oxynitride doped with carbon, amorphous carbon material, silicon carbide, other nitride materials, other carbide materials, or combination thereof, and has an etching selectivity different from those of the dielectric materials for forming the air gaps **32** and the dummy features **24**. Other materials suitable for the passivation layer **61** are within the contemplated scope of the disclosure.

[0044] Referring to FIG. **1** and the example illustrated in FIGS. **14**A and **14**B, the method **100** then proceeds to step **113**, where the passivation layer is patterned. FIG. **14**A illustrates a configuration subsequent to that shown in FIG. **13**A, and FIG. **14**B is a schematic perspective view of a portion of the configuration shown in FIG. **14**A. The passivation layer **61** is patterned such that a plurality of openings **611** in an array arrangement are formed in the passivation layer **61**. In some embodiments, the process for forming an array of the openings **611** in the passivation layer **61** is the same or similar to that for forming the openings **231** in the dummy layer **23** described above, and details thereof are omitted for the sake of brevity. The openings **611** in the array arrangement extend through the passivation layer **61**, and are arranged in position corresponding to the dummy pillars **24**.

[0045] Referring to FIG. **1** and the example illustrated in FIGS. **15**A and **15**B, the method **100** then proceeds to step **114**, where the second elongated slots are reopened. FIG. **15**A illustrates a configuration subsequent to that shown in FIG. **14**A, and FIG. **15**B is a schematic perspective view of a portion of the configuration shown in FIG. **15**A. The dummy pillars **24** shown in FIG. **14**A are removed using a suitable etching process, for example, but not limited to, a wet etching process, a dry etching process, or the like, through the openings **611** of the passivation layer **61** shown in FIG. **14**A, such that the second elongated slots **225** are reopened.

[0046] Referring to FIG. **1** and the example illustrated in FIGS. **16**A and **16**B, the method **100** then proceeds to step **115**, where a plurality of second conductive lines are formed. FIG. **16**A illustrates a configuration subsequent to that shown in FIG. **15**A, and FIG. **16**B is a schematic perspective view of a portion of the configuration shown in FIG. **16**A. A conductive material is filled into the second elongated slots **225** shown in FIG. **15**A using a suitable deposition technique, for example, but not limited to, CVD, PECVD, or the like. Other suitable techniques for filling the conductive material are within the contemplated scope of the present disclosure. The conductive material may include aluminum, zirconium, titanium, tungsten, tantalum, ruthenium, palladium, platinum, cobalt, nickel, or the like, or alloys thereof. Other conductive materials suitable for filling the second elongated slots **225** are within the contemplated scope of the present disclosure. After filling of the conductive material, a planarization process, such as CMP, is performed to remove an excess of the conductive material above the passivation layer **61** and to form a plurality of second conductive lines **71** in an array arrangement. Other suitable processes for planarizing the conductive material are within the contemplated scope of the disclosure. The second conductive lines **71** in the array arrangement are elongated in the Z direction to terminate the etch stop layer **21**. An array of the second conductive lines **71** includes a plurality of columns of the second conductive lines **71** spaced apart from each other in the X direction. The second conductive lines **71** in each of the columns are spaced apart from each other in the Y direction. Accordingly, an embodiment of the semiconductor device **200** of the present disclosure is obtained.

[0047] Referring to the examples illustrated in FIGS. **16**A and **17**A to **17**C, the semiconductor device **200** includes a three-dimensional (3D) memory structure **10** disposed over the substrate **1**. The 3D memory structure **10** includes a plurality of memory cells **11** arranged in the three directions (i.e., the X, Y, and Z directions) which are transverse to one another. In some embodiments, the three directions are perpendicular to one another. In some embodiments, the 3D memory structure **10** is located in the back-end of line (BEOL), while in certain embodiments, the 3D memory structure **10** may be located in the front-end of line (FEOL).

[0048] The 3D memory structure **10** includes a plurality of the first conductive lines **41**, a plurality of the second conductive lines **71**, a plurality of the memory elements **51**, and a plurality of the channel elements **311**. As described above, the first conductive lines **41** are elongated in the Y direction, and are disposed in an array arrangement. An array of the first conductive lines **41** includes a plurality of columns of the first conductive lines **41** spaced apart from each other in the X direction, and the first conductive lines **41** in each of the columns are spaced apart from each other in the Z direction. Each of the first conductive lines **41** serves as a word line. The second conductive lines **71** are elongated in the Z direction, and are disposed in an array arrangement. An array of the second conductive lines **71** includes a plurality of columns of the second conductive lines **71** spaced apart from each other in the X direction, and the second conductive lines **71** in each of the columns are spaced apart from each other in the Y direction. Each of the second conductive lines **71** serves as a source line or a bit line. The memory elements **51** are elongated in the Y direction, and are disposed in an array arrangement. An array of the memory elements **51** includes a plurality of columns of the memory elements **51** spaced apart from each other in the X direction, and the memory elements **51** in each of the columns are spaced part from each other in the Z direction. The channel elements **311** are disposed in a 3D array arrangement. A 3D array of the channel elements **311** includes a plurality of 2D arrays of the channel elements **311** spaced apart from each other in the X direction. Each of the 2D arrays of the channel elements **311** includes a plurality of columns of the channel elements **311** spaced apart from each other in the Z direction, and the channel elements **311** in each of the columns of each of the 2D arrays of the channel elements **311** are spaced apart from each other in the Y direction.
[0049] Each of the channel elements **311** is disposed between two corresponding ones of the first conductive lines **41** which are spaced apart from each other in the Z direction, and is formed with an air gap **32** therein. Two corresponding ones of the memory elements **51** are disposed at two opposite sides of the each of the channel elements **311** in the Z direction, such that the two corresponding ones of the memory elements **51** are disposed to surround the two corresponding ones of the first conductive lines **41**, respectively, and to separate each of the two corresponding ones of the first conductive lines **41** from the each of the channel elements **311**. Two corresponding ones of the second conductive lines **71** are disposed at two opposite sides of the each of the channel elements **311** in the X direction, such that each of the two corresponding ones of the second conductive lines **71** is in contact with the each of the channel elements **311** and the two corresponding ones of the memory elements **51**.
[0050] The each of the channel elements **311** includes an upper channel portion **311***a* and a lower channel portion **311***b*, which are spaced apart from each other by the air gap **32** and which are in contact with the two corresponding ones of the memory elements **51**, respectively, and a first side channel portion **311***c* and a second side channel portion **311***d*, which are spaced apart from each other by the air gap **32** and which are in contact with the two corresponding ones of the second conductive lines **71**, respectively.
[0051] An upper one of the two corresponding ones of the memory elements **51** includes a lower memory portion **51***a* in contact with the each of the channel elements **311**, an upper memory portion **51***b* spaced apart from the lower memory portion **51***a* by an upper one of the two corresponding ones of the first conductive lines **41**, and a first side memory portion **51***c* and a second side memory portion **51***d* which are spaced apart from each other by the upper one of the two corresponding ones of the first conductive lines **41** and which are in contact with the two corresponding ones of the second conductive lines **71**, respectively.
[0052] A lower one of the two corresponding ones of the memory elements **51** includes an upper memory portion **51***b* in contact with the each of the channel elements **311**, a lower memory portion **51***a* spaced apart from the upper memory portion **51***b* by a lower one of the two corresponding ones of the first conductive lines **41**, and a first side memory portion **51***c* and a second side memory portion **51***d* which are spaced apart from each other by the lower one of the two corresponding ones

of the first conductive lines **41** and which are in contact with the two corresponding ones of the second conductive lines **71**, respectively.

[0053] In addition, referring to the examples illustrated in FIGS. **18**A to **18**C, each of the first conductive lines **41** is disposed between two corresponding ones of the channel elements **311** which are spaced apart from each other in the Z direction. Each of the memory elements **51** is disposed to surround the each of the first conductive lines **41** and to separate each of the two corresponding ones of the channel elements **311** from the each of the first conductive lines **41**. Two corresponding ones of the second conductive lines **71** are disposed at two opposite sides of the each of the memory elements **51** in the X direction, such that each of the two corresponding ones of the second conductive lines **71** is in contact with the each of the memory elements **51** and the two corresponding ones of the channel elements **311**.

[0054] Each of the memory elements **51** includes a lower memory portion **51***a* in contact with a lower one of the two corresponding ones of the channel elements **311**, an upper memory portion **51***b* which is spaced apart from the lower memory portion **51***a* by the each of the first conductive lines **41** and which is in contact with an upper one of the two corresponding ones of the channel elements **311**, and a first side memory portion **51***c* and a second side memory portion **51***d* which are spaced apart from each other by the each of the first conductive lines **41** and which are in contact with the two corresponding ones of the second conductive lines **71**, respectively.

[0055] An upper one of the two corresponding ones of the channel elements **311** includes an upper channel portion **311***a*, a lower channel portion **311***b* spaced apart from the upper channel portion **311***a* by the air gap **32** and in contact with the each of the memory elements **51**, and a first side channel portion **311***c* and a second side channel portion **311***d*, which are spaced apart from each other by the air gap **32** and which are in contact with the two corresponding ones of the second conductive lines **71**, respectively.

[0056] A lower of the two corresponding ones of the channel elements **311** includes an upper channel portion **311***a* in contact with the each of the memory elements **51**, a lower channel portion **311***b* spaced apart from the upper channel portion **311***a* by the air gap **32**, and a first side channel portion **311***c* and a second side channel portion **311***d*, which are spaced apart from each other by the air gap **32** and which are in contact with the two corresponding ones of the second conductive lines **71**, respectively.

[0057] Referring to the examples illustrated in FIGS. **18**B, **19**, and **20**, the air gaps **32** formed in the channel elements **311** may have various geometric shapes in cross-section, such as a circular-like shape shown in FIG. **18**B, a rectangular-like shape shown in FIGS. **19**, **20**, or the like. In addition, in some embodiments, the dielectric material **25** filled for forming the air gaps **32** may be deposited on inner surfaces of the channel elements **311** to form dielectric layers on the inner surfaces of the channel elements **311** so as to confine the air gaps **32** as shown in FIGS. **18**B and **19**. In some alternative embodiments, the air gaps **32** are directly confined by the channel elements **311** without formation of the dielectric layers on the inner surfaces of the channel elements **311** as shown in FIG. **20**.

[0058] Referring to the examples illustrated in FIGS. **21** and **22**, the air gaps **32** may be additionally formed in portions of the dielectric material **25** filled among the memory elements **51**. The air gaps **32** formed in the portions of the dielectric material **25** filled among the memory elements **51** may also have various geometric shapes in cross-section, such as a circular-like shape shown in FIG. **21**, a rectangular-like shape shown in FIG. **22**, or the like.

[0059] Referring to the example shown in FIG. **23**, each of the memory cells **11** includes portions of three of the second conductive lines **71**, portions of two of the first conductive lines **41** disposed to alternate with the three of the second conductive lines **71**, portions of two of the memory elements **51** respectively surrounding the two of the first conductive lines **41**, four of the channel elements **311** in an arrangement of a two by two array. Two columns of the channel elements **311** are spaced apart from each other by a middle one of the three of the second conductive lines **71**.

The channel elements **311** in each of the two columns are spaced apart from each other in the Z direction, and are in contact with a corresponding one of the two of the memory elements **51**. The middle one of the three of the second conductive lines **71** serves as a common source line (C-SL), and remaining two ones of the three of the second conductive lines **71** serve as bit lines (BL**1**, BL**2**), respectively. The two of the first conductive lines **41** serve as word lines (WL**1**, WL**2**), respectively. When the data read out in the word line (WL**1**) is 1 with the application of a suitable voltage, the data read out in the word line (WL**2**) is 0 without the application of a voltage, the data read out in the bit line (BL**1**) is 1 with the application of a suitable voltage, the data read out in the bit line (BL**2**) is 0 without the application of a voltage, and the common source line (C-SL) is grounded, channels (C**1**, i.e., the channel portions of the channel elements **311** proximate to the word line (WL**1**)) are turned on to permit currents to flow from the bit line (BL**1**) to the common source line (C-SL) through the channels (C**1**). When the data read out in the word line (WL**1**) is 0 without the application of a voltage, the data read out in the word line (WL**2**) is 1 with the application of a suitable voltage, the data read out in the bit line (BL**1**) is 0 without the application of a voltage, the data read out in the bit line (BL**2**) is 1 with the application of a suitable voltage, and the common source line (C-SL) is grounded, channels (C**2**, i.e., the channel portions of the channel elements **311** proximate to the word line (WL**2**)) are turned on to permit currents to flow from the bit line (BL**2**) to the common source line (C-SL) through the channels (C**2**).
[0060] In the semiconductor device of the present disclosure, the channel elements of the 3D memory structure of the semiconductor device are formed with the air gaps therein. Therefore, a capacitance between two adjacent ones of the word lines disposed at two opposite sides of each of the channel elements can be reduced significantly. In addition, the air gaps may be formed in the dielectric material filled among the second conductive lines serving as the source/bit lines and among the memory elements. Therefore, the total capacitance produced by the 3D memory structure can be further reduced.
[0061] In accordance with some embodiments of the present disclosure, a semiconductor device includes a substrate and a memory structure disposed over the substrate. The memory structure includes a pair of first conductive lines, a channel element disposed between the pair of the first conductive lines and formed with an air gap therein, a first memory element disposed to separate one of the pair of the first conductive lines from the channel element, and a second memory element disposed to separate the other one of the pair of the first conductive lines from the channel element.
[0062] In accordance with some embodiments of the present disclosure, the memory structure further includes a pair of second conductive lines disposed transversely relative to the pair of the first conductive lines and at two opposite sides of the channel element such that each of the second conductive lines is in contact with the channel element and the first and second memory elements.
[0063] In accordance with some embodiments of the present disclosure, the channel element includes an upper channel portion disposed to be in contact with the first memory element, a lower channel portion disposed to be in contact with the second memory element, a first side channel portion disposed to be in contact with one of the second conductive lines, and a second side channel portion disposed to be in contact with the other one of the second conductive lines.
[0064] In accordance with some embodiments of the present disclosure, the first memory element surrounds one of the first conductive lines, and includes a lower memory portion disposed to be in contact with the channel element, an upper memory portion disposed on the one of the first conductive lines and opposite to the lower memory portion, a first side memory portion disposed to be in contact with one of the second conductive lines, and a second side memory portion disposed to be in contact with the other one of the second conductive lines.
[0065] In accordance with some embodiments of the present disclosure, the second memory element surrounds the other one of the first conductive lines, and includes an upper memory portion disposed to be in contact with the channel element, a lower memory portion disposed on

the other one of the first conductive lines and opposite to the upper memory portion, a first side memory portion disposed to be in contact with one of the second conductive lines, and a second side memory portion disposed to be in contact with the other one of the second conductive lines.

[0066] In accordance with some embodiments of the present disclosure, the semiconductor device further includes a dielectric layer formed on an inner surface of the channel element to confine the air gap.

[0067] In accordance with some embodiments of the present disclosure, each of the first conductive lines serves as a word line.

[0068] In accordance with some embodiments of the present disclosure, one of the second conductive lines serves as a source line and the other one of the second conductive lines serves as a bit line.

[0069] In accordance with some embodiments of the present disclosure, a semiconductor device includes a substrate and a memory structure disposed over the substrate. The memory structure includes a first channel element and a second chancel element, each of which is formed with an air gap therein, a first conductive line disposed between the first and second channel elements, and a memory element disposed to separate the first conductive line from the first and second channel elements.

[0070] In accordance with some embodiments of the present disclosure, the memory structure further includes a pair of second conductive lines disposed transversely relative to the first conductive lines and at two opposite sides of the memory element such that each of the second conductive lines is in contact with the first and second channel elements and the memory element.

[0071] In accordance with some embodiments of the present disclosure, the first channel element includes a lower channel portion disposed to be in contact with the memory element, an upper channel portion spaced apart from the lower chancel portion by the air gap, a first side channel portion disposed to be in contact with one of the second conductive lines, and a second side channel portion disposed to be in contact with the other one of the second conductive lines.

[0072] In accordance with some embodiments of the present disclosure, the second channel element includes an upper channel portion disposed to be in contact with the memory element, a lower channel portion spaced apart from the upper channel portion by the air gap, a first side channel region disposed to be in contact with one of the second conductive lines, and a second side channel portion disposed to be in contact with the other one of the second conductive lines.

[0073] In accordance with some embodiments of the present disclosure, the memory element surrounds the first conductive line, and includes an upper memory portion disposed to be in contact with the first channel element, a lower memory portion disposed to be in contact with the second channel element, a first side memory portion disposed to be in contact with one of the second conductive lines, and a second side memory portion disposed to be in contact with the other one of the second conductive lines.

[0074] In accordance with some embodiments of the present disclosure, the first conductive line serves as a word line.

[0075] In accordance with some embodiments of the present disclosure, one of the second conductive lines serves as a source line and the other one of the second conductive lines serves as a bit line.

[0076] In accordance with some embodiments of the present disclosure, the semiconductor device further includes a dielectric layer formed on an inner surface of each of the first and second channel elements to confine the air gap.

[0077] In accordance with some embodiments of the present disclosure, a method for manufacturing a semiconductor device includes: forming a first dielectric pillar having a first wall surface, and a second dielectric pillar having a second wall surface which is spaced apart from the first wall surface; forming a first dummy pillar and a second dummy pillar which are spaced apart from each other and which extend transversely relative to the first and second dielectric pillars,

such that the first and second dummy pillars are interconnected with each other through the first and second dielectric pillars, the first dummy pillar having a third wall surface extending to interconnect the first and second dielectric pillars, the second dummy pillar having a fourth wall surface spaced apart from the third wall surface and extending to interconnect the first and second dielectric pillars; conformally forming a channel feature which includes a first channel region disposed between and in contact with the first and second dummy pillars, and a second channel region exposed beyond the first and second dummy pillars; removing the first and second dielectric pillars such that the first and second dummy pillars are interconnected with each other through the channel feature; forming a pair of first conductive lines, a first memory element, and a second memory element, the pair of the first conductive lines being disposed transversely relative to the first and second dummy pillars and opposite to each other relative to the channel feature, each of the first and second memory elements including a first memory portion in contact with the channel feature, a second memory portion spaced apart from the first memory portion by a corresponding one of the first conductive lines, and a third memory portion and a fourth memory portion, each of which extends to interconnect the first and second memory portions and to contact a corresponding one of the first and second dummy pillars and the corresponding one of the first conductive lines; and removing at least a portion of the second channel region to form a channel element having an air gap therein.
[0078] In accordance with some embodiments of the present disclosure, the method for manufacturing a semiconductor device further includes: replacing the first and second dummy pillars with a conductive material to form a pair of second conductive lines transversely relative to the first conductive lines.
[0079] In accordance with some embodiments of the present disclosure, the method for manufacturing a semiconductor device further includes: forming a dielectric layer on an inner surface of the channel element to confine the air gap.
[0080] In accordance with some embodiments of the present disclosure, the dielectric layer is formed by filling a dielectric material into an elongated slot passing through the channel element.
[0081] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes or structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device comprising: a substrate; and a memory structure disposed over the substrate and including: a pair of first conductive lines extending in a first direction, a channel element disposed between the pair of first conductive lines, a first air gap located in the channel element, a first memory element extending in the first direction and disposed to separate one of the first conductive lines in the pair of first conductive lines from the channel element, and a second memory element extending in the first direction and disposed to separate the other one of the first conductive lines in the pair of first conductive lines from the channel element.
2. The semiconductor device according to claim 1, wherein the memory structure further includes a pair of second conductive lines extending in a second direction different from the first direction and disposed at two opposite sides of the channel element, each of the second conductive lines being in contact with the channel element and the first memory element and the second memory element.
3. The semiconductor device according to claim 1, the memory structure further includes a second

air gap, and the first memory element and the second memory element are spaced apart from each other by the second air gap in a second direction different from the first direction.

4. The semiconductor device according to claim 3, wherein the memory structure further includes a dielectric material filled between the first memory element and the second memory element, and the second air gap is located in the dielectric material.

5. The semiconductor device according to claim 2, wherein the channel element includes: an upper channel portion, a lower channel portion spaced apart from the the upper channel portion by the first air gap in the second direction, a first side channel portion connected to the upper channel portion and the lower channel portion, and a second side channel portion connected to the upper channel portion and the lower channel portion, and spaced apart from the first side channel portion by the first air gap in a third direction different from the first direction and the second direction.

6. The semiconductor device according to claim 1, wherein the first memory element includes: a lower memory portion in contact with the channel element, an upper memory portion spaced apart from the lower memory portion by the one of the first conductive lines in the pair of first conductive lines in a second direction transvers to the first direction, a first side memory portion connected to the lower memory portion and the upper memory portion, and a second side memory portion connected to the lower memory portion and the upper memory portion, and spaced apart from the first side memory portion by the one of the first conductive lines in the pair of first conductive lines in a third direction different from the first direction and the second direction.

7. The semiconductor device according to claim 6, wherein the lower memory portion includes a first memory part in contact with the channel element, and a second memory part connected to the first memory part and protruding from the channel element in the first direction.

8. The semiconductor device according to claim 1, wherein the second memory element includes: an upper memory portion in contact with the channel element, a lower memory portion spaced apart from the upper memory portion by the other of the first conductive lines in the pair of first conductive lines in a second direction transvers to the first direction, a first side memory portion connected to the lower memory portion and the upper memory portion, and a second side memory portion connected to the lower memory portion and the upper memory portion, and spaced apart from the first side memory portion by the other of the first conductive lines in the pair of first conductive lines in a third direction different from the first direction and the second direction.

9. The semiconductor device according to claim 8, wherein the upper memory portion includes a first memory part in contact with the channel element, and a second memory part connected to the first memory part and protruding from the channel element in the first direction.

10. A semiconductor device comprising: a substrate; and a memory structure disposed over the substrate and including: a first channel element and a second channel element, an air gap located in at least one of the first channel element and the second channel element, a first conductive line extending in a first direction and disposed between the first channel element and the second channel element, a memory element disposed between the first conductive line and at least one of the first channel element and the second channel element, and a second conductive line extending in a second direction different from the first direction and connected to the first channel element, the second channel element, and the memory element.

11. The semiconductor device according to claim 10, wherein the memory element includes: a lower memory portion in contact with the first channel element, an upper memory portion in contact with the second channel element, a first side memory portion connected to the lower memory portion and the upper memory portion, and a second side memory portion connected to the lower memory portion and the upper memory portion, and spaced apart from the first side memory portion by the first conductive line in a third direction different from the first direction and the second direction.

12. The semiconductor device according to claim 11, wherein the first channel element and the first conductive line are spaced apart from each other by the lower memory portion.

13. The semiconductor device according to claim 11, wherein the second channel element and the first conductive line are spaced apart from each other by the upper memory portion.
14. The semiconductor device according to claim 11, wherein the second conductive line and the first conductive line are spaced apart from each other by one of the first side memory portion and a second side memory portion.
15. The semiconductor device according to claim 11, wherein the lower memory portion includes a first memory part in contact with the first channel element, and a second memory part connected to the first memory part and protruding from the first channel element in the first direction.
16. The semiconductor device according to claim 11, wherein the upper memory portion includes a first memory part in contact with the second channel element, and a second memory part connected to the first memory part and protruding from the second channel element in the first direction.
17. A method for manufacturing a semiconductor device, comprising: forming a channel element which includes a first channel portion, a second channel portion disposed below the first channel portion, and two side channel portions connected to the first channel portion and the second channel portion; forming a first memory element and a second memory element extending in a first direction, wherein the first memory element is disposed on the first channel portion and the second memory element is disposed under the second channel portion; forming a pair of first conductive lines extending in the first direction, wherein the pair of first conductive lines contact the first memory element and the second memory element, respective; and forming a pair of second conductive lines extending in a second direction different from the first direction, wherein the pair of second conductive lines contact the channel element, the first memory element, and the second memory element.
18. The method according to claim 17, further comprising: forming a first air gap in the channel element.
19. The method according to claim 18, further comprising: forming a second air gap between the first memory element and the second memory element.
20. The method according to claim 19, further comprising: filing a dielectric material in the channel element and between the first memory element and the second memory element, wherein the first air gap and the second air gap are formed in the dielectric material.